

Summary Notes on Non-linear Growth and Competitive Coexistence

CNRS - Station d'écologie théorique et expérimentale

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1 Summary

For anyone who happens to read this, these notes present a summary of my work done during an internship at CNRS — they are neither thoroughly detailed nor written with the intent of being graded or published. That being said, I enjoyed working with JF Arnoldi and hope he will find the work we did of some use, but if he doesn't that's fine too. I believe that, even if only as a training exercise, questioning the pillars of a subject can be a valuable experience.

These notes focus on the ramifications of non-linear growth — particularly sub-linear — and how it pertains to competitive coexistence. Non-linear growth is by no means a revolutionary idea in ecology, but it is typically assumed that populations grow logistically, and even if this isn't strictly true, it works as a simple, effective approximation that can be used to show certain phenomena. For example, by considering a consumer-resource model with species experiencing logistic growth, it is easy to show prove that competitive exclusion will occur.

However, once non-linear growth is considered, competitive exclusion is not a guarantee; sub-linear growth theoretically allows for species possessing higher R -star values to survive at low abundances. This is also seen in the simulations created, with species surviving at higher abundances than expected (although the fitter species is still the dominant species). The effects of supra-linear growth were also considered. Interestingly, the order species are introduced to the environment makes a difference to the stable solution for supra-linear growth models, but this wasn't investigated as deeply because there is less evidence to support supra-linear growth as a common occurrence.

N.B. unless otherwise specified, information and equations used are taken from Michel Loreau's book, "From Populations to Ecosystems: Theoretical Foundations for a New Ecological Synthesis" [1].

2 Background Theory and Motivation For Investigating Non-Linear Growth

A core tenet in ecology and population dynamics is logistic growth. This is the idea that populations grow exponentially when they are small and then plateau as resources become limited. This is outlined mathematically in the 2024 paper [2], and discussed in more detail in their supplementary materials; for a species with a time-dependent abundance $x(t)$, its growth can be modelled as the differential equation

$$\frac{1}{x} \frac{dN}{dt} = \frac{\alpha}{\theta} \left[1 - \left(\frac{x}{K} \right)^\theta \right], \quad (1)$$

where K is the carrying capacity — the maximum abundance for a particular system — and α is a parameter that can be written in terms of the intrinsic growth rate — the growth rate when the abundance is far from the carrying capacity. θ indicates the degree of non-linearity: the system is linear and hence corresponds to logistic growth when $\theta = 1$, and is otherwise non-linear. The visual differences between exponential and logistic growth can be seen in figure 1. While a population growing exponentially grows at an ever increasing rate, a population growing logistically will have a symmetric growth curve, with the growth rate peaking at an abundance equal to $K/2$ and plateauing at an abundance equal to K .

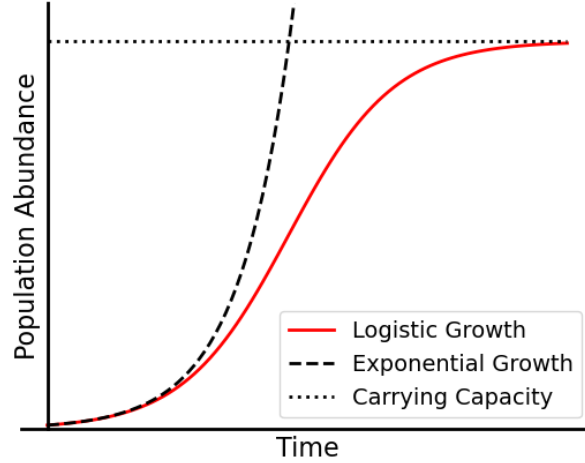


Figure 1: Difference between exponential growth (the black, dashed line) and logistic growth (the solid, red line). Notice that the logistic growth levels off at the carrying capacity (the black, dotted line)

Non-linear growth, when $\theta \neq 1$ in equation 1, is similar to logistic growth in that it grows quickly at low abundances, and then plateaus as the carrying capacity is reached. The difference between linear and non-linear growth lies in the symmetry of the growth curves. Whereas the growth rate peaks at $K/2$ for logistic growth and its curve exhibits rotational symmetry about this point, non-linear growth disrupts this symmetry. When compared to logistic growth, general supra-linear growth, $\theta > 1$, results in a lower level of growth at lower abundances,

followed by a sharp levelling off as the abundance approaches the carrying capacity. In contrast, sub-linear growth results in a higher level of growth at low abundances but a more gradual levelling off as the abundance approaches the carrying capacity. This is shown in figure 2 for $\theta = -0.5, \theta = 0.5$ and $\theta = 1.5$, where a logistic growth curve ($\theta = 1$) is included for reference. It should also be noted that while, according to equation 1, $\theta = 0$ would appear to be undefined, the supplementary materials for [2] recover Gompertz growth, which is

$$\frac{dx}{dt} = \alpha \ln\left(\frac{K}{x}\right).$$

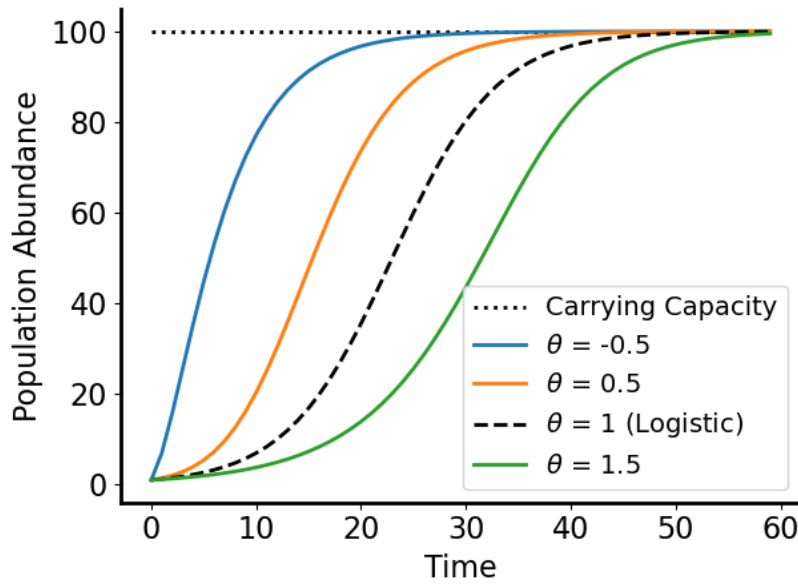


Figure 2: Different types of non-linear growth. Logistic growth is shown for reference as a dotted line in the centre. In comparison to logistic growth, supra-linear growth (shown for $\theta = 1.5$) causes a shift to the right, and sub-linear growth (shown for both $\theta = 0.5$ and $\theta = -0.5$) causes a shift to the left.

If logistic growth is a generally accepted assumption in ecology, what would be the incentive to challenge it? The intuitive answer is that logistic growth is an idealised representation of growth. For a typical consumer-resource model (more detail in section 3), it is assumed that both the consumption and mortality terms of a species are independent of density, when in reality there are many reasons to expect this would not be exactly so: mortality could increase at high densities due to increased disease transmission rates or consumption could fall due to increased intraspecific competition. In addition, a recent article [2] was published in March 2024 that summarised the large-scale analysis of the data on mammals, birds, fish and insects performed in [3]. As shown in figure 3, the growth for most of the groups considered in [3] are sub-linear rather than linear, and although there are concerns about the reliability of the data for large mammals [4], this data does seem to provide evidence that logistic growth is not always the norm in nature.

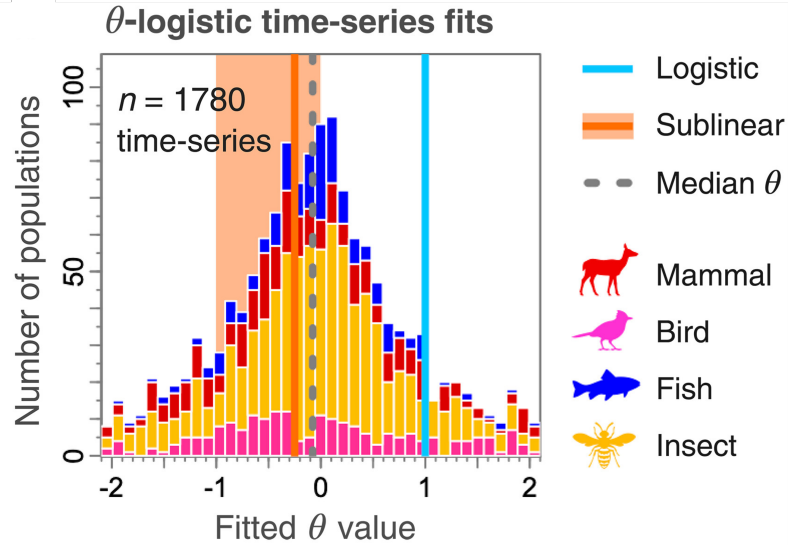


Figure 3: Data, analysed by [3] and summarised by [2], showing a combination of 1780 population-time series for mammals, birds, fish and insects. The median value for θ is -0.08.

It is worth noting that this isn't a new idea, and has been considered many times in theoretical ecology, including a 2009 paper that assumed a general weakly-dependent mortality model [5], and a number of studies on plankton populations from 1992–2001 [6][7][8]; instead we investigate the idea of non-linearity at a general, fundamental level by considering simple changes to a consumer-resource model and the consequences they may have.

3 A Sub-linear Consumer-resource Model

Consider a single-resource consumer-resource model for a species i . Similarly to equation 1, an exponent term ϕ can be used to indicate the degree of non-linearity, but unlike θ in equation 1, ϕ is always positive. A sub-linear term $\langle x_i \rangle^{-\phi} = [1 + (x_i/K)]^{-\phi}$ is used which ensures species i will still behave logistically at low abundance, but that the sub-linear effects will become clear when x_i is on the same order of magnitude as K . The consumer-resource equations for species i can accordingly be written as

$$\frac{1}{x_i} \frac{dx_i}{dt} = \varepsilon c R \langle x \rangle^{-\phi} - m \quad (2a)$$

$$\frac{dR}{dt} = l(R_0 - R) - c R x_i \langle x \rangle^{-\phi}, \quad (2b)$$

where ε is the efficiency for consumers converting resources into new consumers, c is the consumption rate per consumer, m is the baseline mortality rate, R is the current resource level and R_0 is the initial resource level before consumption. Note that, when $-1 < \theta < 0$, equation 1 can be written as

$$\frac{1}{N} \frac{dN}{dt} = \frac{\alpha}{|\theta|} \left[\left(\frac{N}{K} \right)^{\theta} - 1 \right],$$

allowing the consumer-resource model in equation 2 to be considered analogous to equation 1 for the range $-1 < \phi < 1$. The traditional argument for competitive exclusion is that, under logistic growth, species i will have a stable abundance ($dx_i/dt = 0$) when the resource level $R_i^* = m/(\varepsilon c R)$, and any other species such as species k will die off if $R_k^* > R_i^*$. The introduction of the sub-linear term $\langle x \rangle^{-\phi}$ changes the R^* value for a species. For instance, the new stable resource level for species i is $r_i^* = m \langle x \rangle^{\phi} / (\varepsilon c R) = \langle x \rangle^{\phi} R_i^*$. Since $\langle x \rangle^{\phi}$ is always positive, r_i^* will always be equal to or greater than R_i^* . Provided R_k^* is between R_i^* and r_i^* (a range whose size is determined by $r_i^* - R_i^* = (\langle x \rangle^{\phi} - 1) R_i^*$), species k is now able to survive, albeit only when x_i is sufficiently large to push the resource level to r_i^* . A visual representation for how this allows species k to coexist with species i is shown in figure 4. In figure 4, species k can coexist if its R_k^* value is in between R_i^* and r_i^* . If $R_k^* > r_i^*$, species k will die off, and if $R_k^* < r_i^*$, species k will out-compete species i , and species i will die off. A similar argument can be used when considering a supra-linear mortality term,

$$\frac{1}{x_i} \frac{dx_i}{dt} = \varepsilon c R - m \langle x \rangle^{\phi} \quad (3a)$$

$$\frac{dR}{dt} = l(R_0 - R) - c R x_i, \quad (3b)$$

with r_i^* once more being equal to $\langle x \rangle^{\phi} R_i^*$. This is analogous to the range $0 < \theta < 1$ in equation 1, so even though we term it a supra-linear mortality term because the mortality dependence is $x_i \langle x \rangle^{\phi} \approx x_i^{1+\phi}$, this is still sub-linear growth rather than supra-linear growth. The two different sub-linear growth models (equations 2 and 3) are broadly similar, and should have the same shift in R^* values. Nevertheless, there is one key difference: the presence of $\langle x_i \rangle^{-\phi}$ in equation 2b for the sub-linear consumption dependence but not in equation 3b for the supra-linear mortality dependence. One prediction that can be made from this is that the sub-linear consumption model will result in higher abundances at stable solutions ($dx/dt = 0$ and $dR/dt = 0$). This is because the stable resource value for equation 2b, R_c^* , is larger than the stable resource value for equation 3b, R_m^* . Specifically, $R_c^* = \langle x^* \rangle R_m^*$. This higher resource level will result in a higher abundance for each species.

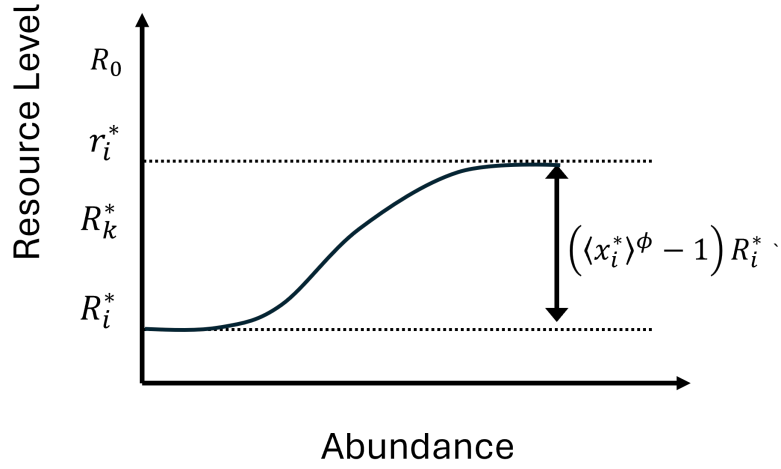


Figure 4: Illustration of how competitive coexistence can stem from non-linear growth. R_i^* shows the resource-level at which species i is stable under logistic growth. For sub-linear growth, the R^* value will increase as abundance increases up until r_i^* growth (roughly shown by the black line), which allows other species (such as species k with R_k^*) to coexist at low abundances in comparison to i .

3.1 Supra-linear Peculiarities

So far, general supra-linear growth (corresponding to $\theta > 1$ in equation 1) has not been considered. This is in part because there are fewer reasons to expect it, and because most of the data in figure 3 shows sub-linear growth. However, there are a non-significant number of data points in figure 3 that have values of θ greater than one, and a strong Allee effect could offer a plausible explanation for how supra-linear could occur.

One peculiar consequence of assuming supra-linear growth concerns the introduction of species into an environment. If species i is fitter and is introduced into an environment where species k is dominant, the abundance of species i will grow and species k will die off. If instead, species i is dominant and species k is introduced, species i will remain dominant and species k will once again die off. Essentially, the order in which species are introduced does not affect the stable solution. However, if supra-linear growth is considered, it does. In contrast to sub-linear growth, $r_i^* = R_i^* \langle x \rangle^{-\phi}$ under supra-linear growth, meaning $r_i^* < R_i^*$. Now imagine that, for species k , $r_i^* < R_k^* < R_i^*$. If species i is introduced into an environment where species k is dominant, species i will die off as, at low abundance, it requires a higher resource level than species k ($R_k^* < R_i^*$). However, if species k is introduced into an environment where species i is already dominant, the resource level will be at r_i^* rather than R_i^* , meaning species k will not be able to grow in the first place (since $r_i^* < R_k^*$). This is illustrated in figure 5.

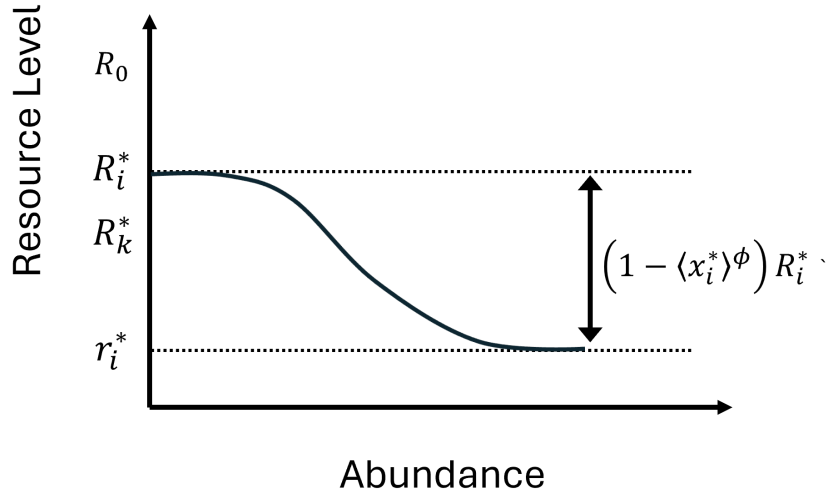


Figure 5: Illustration of supra-linear peculiarity. If $r_i^* < R_k^* < R_i^*$, the order in which species are introduced into the environment determines which species survive. If species k is introduced first, it will survive and species i will die off, but if species i is introduced first it is the other way round. .

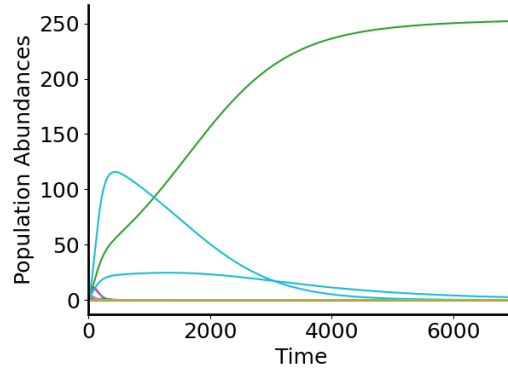
As previously stated, it is harder to understand how significant supra-linear growth would arise than to understand how sub-linear growth would; there is also less evidence for supra-linear growth. This peculiarity could therefore be viewed as further evidence that supra-linear growth is nonsensical, or it could simply be viewed as a peculiar outcome of a certain type of growth — one that may or may not be seen in nature.

4 Sub-linear Growth Simulations

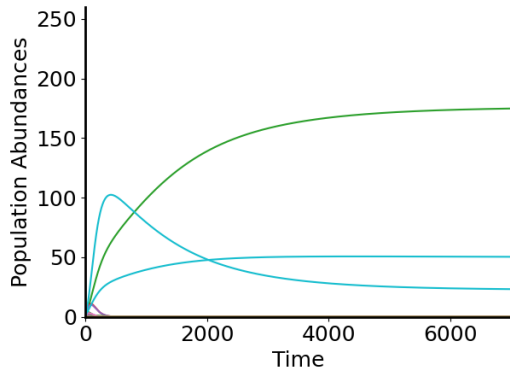
In order to gain a better understanding of the coexistence behaviour that non-linearity may introduce, a number of simulations were carried out in python. These simulations were not analysed in depth as the intention was simply to gain a more intuitive understanding of non-linear behaviour rather than quantifying specific relationships. For the simulations, thirty species were randomly chosen with intrinsic consumption and mortality rates between 0 and 1 and 0 and 0.9 respectively. Other variables were kept constant ($\varepsilon = 0.5$, $l = 0.5$ and $R_0 = 10$) and the system was allowed to evolve for 7000 time steps. The consumption and mortality rates were randomised within their respective ranges until non-linear growth resulted in coexistence of multiple dominant species, values which remained constant for all values of ϕ . I did not investigate the consumption and mortality rate criteria for stable coexistence of multiple dominant species, but this appeared to occur in slightly less than half of the simulation runs. In the other cases, a single dominant species would arise with either a few species surviving at much lower levels or all other species appearing to die off.

The results of these simulations are shown in figure 6 for logistic growth and non-linear growth at both small deviations from linearity ($\phi = 0.05$) and large deviations ($\phi = 0.5$). Additionally, the differences between sub-linear density-dependent consumption terms (shown in figures 6b and 6d) and supra-linear density-dependent mortality terms (shown in figures 6c and 6e) were compared, corresponding to the differences between equations 2 and 3.

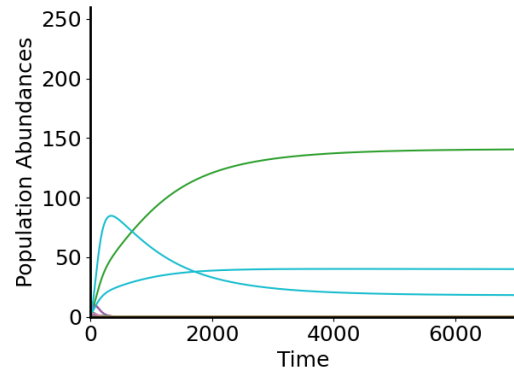
The first noticeable trait is that even figures 6b and 6c with relatively small deviations from linearity ($\phi = 0.05$) produce demonstrate coexistence. However, these figures also appear to suggest that fewer species will be able to coexist than at larger deviations from linearity with only three species surviving; in contrast, many species appear to survive (often at low levels) with a large deviation from linearity ($\phi = 0.5$). Intuitively, this is to be expected as, looking at figure 4, a larger ϕ value results in a larger gap between R_i^* and r_i^* , and it is therefore more likely species will have an R^* value within this range. Two other general trends that are clear are: the higher the ϕ value considered, the shorter the time taken to reach stability; and a consumer-resource model with a supra-linear density-dependent mortality term instead of a sub-linear density-dependent consumption term (i.e. using equation 3 instead of equation 2) results in a lower abundance for each respective species. The latter was predicted in section 3 due to their respective stable resource-level solutions. These trends are also seen for other groups of species that have the same ϕ value as each other, but I have not yet investigated the effect of multiple species with different ϕ values — where each species would have a different degree of non-linearity.



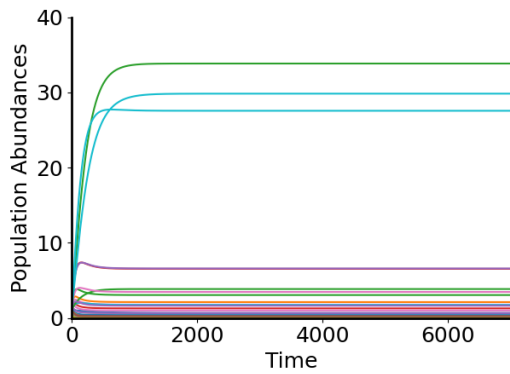
(a) Logistic growth



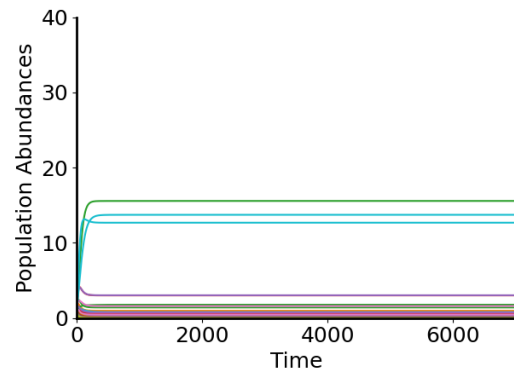
(b) Sub-linear, density-dependent consumption term ($\phi = 0.05$)



(c) Supra-linear, density-dependent mortality term ($\phi = 0.05$)



(d) Sub-linear, density-dependent consumption term ($\phi = 0.5$)



(e) Supra-linear, density-dependent mortality term ($\phi = 0.05$)

Figure 6: Simulations for two types of sub-linear growth: sub-linear density-dependent consumption growth (figures 6b and 6d) and supra-linear density-dependent mortality growth (figures 6c and 6e). Logistic growth (figure 6a) has been included to demonstrate competitive exclusion. For all four figures demonstrating sub-linear growth, competitive coexistence is apparent.

References

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